Lesson 32: Additional Considerations, the Replicability Crisis and Open Science

English Stage 4 (Psychology) · Unit 4: Statistics in Research Reports

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 13.4 Additional Considerations; 13.5 From the "Replicability Crisis" to Open Science Practices; 13.6 Inferential Statistics (Summary).

In this lesson

No statistical decision is guaranteed to be right. You will learn about Type I and Type II errors, statistical power, criticisms of null hypothesis testing, psychology's "replicability crisis" and the open science practices that respond to it. You will practise the language of concession and hedging, and complete the results section of your project.

Learning goals

After this lesson you should be able to:

- Define Type I and Type II errors and explain how α, power and the file drawer problem affect them.
- Use the book's power table to plan a sample size.
- Summarize the criticisms of null hypothesis testing and the recommended responses.
- Explain the replicability crisis, questionable research practices and open science, using concession and hedging.

Key language

Term or phrase	Meaning	Example
Type I error	rejecting the null hypothesis when it is true	concluding there is a relationship when there is not
Type II error	failing to reject the null hypothesis when it is false	missing a real relationship with too small a sample
file drawer problem	statistically significant results are more likely to be submitted and published than nonsignificant results	non-significant results put away "in a file drawer"
p-hacking	making research decisions until the results give a desirable *p* value	arbitrarily removing outliers
statistical power	the probability of rejecting the null hypothesis for a given sample size and expected relationship strength	Power of .80 is adequate.

Term or phrase	Meaning	Example
confidence interval	a range computed so that, usually 95% of the time, the population parameter lies within it	160 to 240 calories
replicability crisis	the inability of researchers to replicate earlier research findings	only 36 of 100 replications were significant
open science practices	practices that make research transparent and open	pre-registration, open data, open materials
although ..., ...	concession: admits one point before the main claim	Although not widely adopted, this rejection ... has continued the conversation.
does not necessarily	a hedge that limits a conclusion	A failure to replicate a result by itself does not necessarily discredit the original study.

Rules and explanations

The ideas

Errors. Two decisions (reject, retain) × two states of the world (the null hypothesis is false, true) give two correct decisions and two errors.

- **Type I error:** rejecting a true null hypothesis. If the null hypothesis is true and α = .05, this happens 5% of the time (α is the "Type I error rate").
- **Type II error:** retaining a false null hypothesis, mainly because the design lacks power (for example, a small sample).
- α = .01 reduces Type I errors but increases Type II errors; α = .10 does the opposite; .05 keeps both acceptable. Any single study may be an error, so researchers **replicate**.

The file drawer problem (Rosenthal, 1979). Significant results are published; non-significant ones stay in the "file drawer". So the literature contains more Type I errors than expected and overstates effects (if the true correlation is +.10, studies may find −.10 to +.40, but mainly the stronger ones are published). Responses: **registered reports** (journals evaluate a study without knowing its results), sharing non-significant results, and the *Journal of Articles in Support of the Null Hypothesis*. Simonsohn, Nelson and Simmons (2014) also accused researchers of ***p*-hacking**: removing outliers arbitrarily, choosing which dependent variables to report, presenting only significant results.

Statistical power = 1 − the probability of a Type II error. With $N = 50$ and $r = +.30$, power is .59, so the Type II risk is .41. Aim for .80 **before** collecting data. Sample sizes for power of .80 (two-tailed):

Relationship strength	Independent-samples *t* test	Test of Pearson's *r*
Strong ($d = 0.80$, $r = .50$)	52	28
Medium ($d = 0.50$, $r = .30$)	128	84

Relationship strength	Independent-samples *t* test	Test of Pearson's *r*
Weak (*d* = 0.20, *r* = .10)	788	782

With 20 participants per condition and *d* = 0.50, power is only .34. To increase power, strengthen the relationship (a stronger manipulation, better control, a within-subjects design) or, usually, enlarge the sample.

Criticisms of null hypothesis testing

- *Misunderstood: p* is not the probability that the null hypothesis is true, and 1 – *p* is not the probability of replication (in Oakes's study, 60% of researchers thought *p* = .01 meant a 99% chance of replicating).
- *Illogical:* a rigid line makes *p* = .04 publishable and *p* = .06 "not significant".
- *Uninformative:* rejection only says the relationship is not exactly zero. Defenders (Abelson, 1995) answer that it gives a principled way to show that results are not mere chance. In 2015, the editors of *Basic and Applied Social Psychology* banned null hypothesis testing (they remove *p* values before publication); cite them as Trafimow and Marks (2015), not "Tramimow" as the book misprints.
- **Recommended:** an **effect size** with every test; **confidence intervals** (*M* = 200 calories, 95% CI 160 to 240, so the mean differs significantly from 250, which lies outside); or **Bayesian statistics**. Null hypothesis testing with effect sizes and confidence intervals remains dominant.

The replicability crisis. The Many Labs project failed to replicate Schnall et al.'s (2008) finding that hand washing makes moral transgressions seem less wrong. In the **Reproducibility Project**, over 270 psychologists repeated 100 published experiments: 97 originals were significant, but only 36 replications, with effect sizes on average half as large. A failed replication does not necessarily discredit the original (power, populations, procedures or moderating variables may differ), but some see **questionable research practices**: selective deletion of outliers, selective reporting ("cherry-picking"), **HARKing** (hypothesizing after the results are known), *p*-hacking (checking significance before deciding to collect more data) and, as fraud, fabrication (Stapel).

Responses: sufficient power; publishing null and significant findings; methods described in enough detail to replicate; high-quality replications; and **open science**, such as badges in *Psychological Science* for pre-registered hypotheses and analysis plans, open materials and open data, and the Transparency and Openness Promotion (TOP) guidelines adopted by more than 500 journals.

Language focus

Concession and hedging in critical discussion. The book admits the other side before making its point, and it limits its claims.

- Concession: "**Although** they did not propose a better statistical test ..., the editors emphasized the importance of descriptive statistics and effect sizes." / "**To be fair**, many researchers have come to the defense of null hypothesis testing." / "**Even those who** defend null hypothesis testing recognize many of the problems with it."

- Hedging: "the published literature **probably** contains a higher proportion of Type I errors"; "**is likely to** overstate the strength of that relationship"; "**does not necessarily** discredit the original study".

> **Note for Arabic speakers** - Do not add *but* after *although*: "Although the result was significant, the effect was small" (not "Although …, but …").

Practice

Part 1: Reading questions

1. Why do Type I errors occur, and why do Type II errors mainly occur?
2. How does the file drawer problem distort the literature, and how do registered reports help?
3. A study has power of .59. What is the probability of a Type II error?
4. Using the power table, how many participants are needed for a medium effect with an independent-samples *t* test, and with a test of *r*?
5. What did the Reproducibility Project find?
6. What is HARKing, and why is it a questionable practice?

Part 2: Exercises from the book (section 13.6)

1. Discussion: A researcher compares the effectiveness of two forms of psychotherapy for social phobia using an independent-samples *t*-test. (a) Explain what it would mean for the researcher to commit a Type I error. (b) Explain what it would mean for the researcher to commit a Type II error.
2. Discussion: What do you think are some of the key benefits of the adoption of open science practices such as pre-registration and the sharing of raw data and research materials? Can you identify any drawbacks of these practices?
3. Practice: Read the online article "Science isn't broken: It's just a hell of a lot harder than we give it credit for" and use the interactive tool entitled "Hack your way to scientific glory" in order to better understand the data malpractice of "*p*-hacking." (Use the archived copies in Watch online, and write three sentences on what you noticed.)

(The book's Type I and Type II problems from Carnegie Mellon's Open Learning Initiative now require a login; ask your tutor for similar problems instead.)

Part 3: Language exercise. Rewrite each overstatement with the word in brackets.

1. "The failure to replicate proves that the original study was wrong." (*necessarily*)
2. "Null hypothesis testing is criticized. It is still the dominant approach." (*although*)
3. "The published literature overstates the strength of relationships." (*likely*)
4. "Pre-registration solves the replicability crisis." (*may*)

Speak with your tutor

- "Let me explain Type I and Type II errors with the book's pregnancy-test cartoon (the test's result versus the truth). Correct my explanation."
- "Play a journal editor who wants to ban *p* values. I will defend null hypothesis testing with effect sizes and confidence intervals, using *although* and *to be fair*."
- "In PSYC 110 Lecture 15 (Morality), Paul Bloom says that Milgram's obedience findings have 'been replicated with many subjects'. Let me explain why replication makes a result more convincing, using Type I errors and the file drawer problem."

Writing task

Your research project: results, part 6 (the complete results section). Combine Lessons 27 to 31 into one results section of 200 to 280 words (**invented data for practice**: nap $M = 22.00$, $SD = 2.54$; wake $M = 17.00$, $SD = 3.09$; $n = 10$ each; $t(18) = 3.95$, $p < .001$, $d = 1.86$; $r(8) = .45$, $p = .20$). Apply this lesson:

- State that the hypotheses, exclusion criterion and analyses were pre-registered (imagine they were), and report the excluded participant.
- Give an effect size with every test; you may add the 95% confidence interval for the difference in means, 2.34 to 7.66 word pairs.
- Label any analysis you did not plan as exploratory.
- Use the power table to explain, with a hedge, why the non-significant correlation may be a Type II error.

If you work on your own topic, revise your own results in the same way.

Checklist:

- Are the planned analyses reported first, and any exploratory analysis labelled?
- Does every test have an effect size, in correct APA form?
- Did you report the exclusion with its criterion?
- Did you avoid "proved", "accept the null hypothesis" and "there is no relationship"?
- Did you hedge your interpretation of the non-significant result?
- Did you use *although* without *but*?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Registered reports, explained by the Center for Open Science: https://cos.io/rr/
- G*Power, a free program for computing statistical power: http://www.gpower.hhu.de
- *Nature* news on the Reproducibility Project's first results: http://www.nature.com/news/first-results-from-psychology-s-largest-reproducibility-test-1.17433

- "Science isn't broken" (FiveThirtyEight has closed; archived copy): https://web.archive.org/web/20240102004641/https://fivethirtyeight.com/features/science-isnt-broken/
- "Hack your way to scientific glory", the interactive *p*-hacking tool (archived copy): https://web.archive.org/web/20240119223724/https://projects.fivethirtyeight.com/p-hacking/

Answer key

Part 1

1. Type I: even with no relationship in the population, sampling error occasionally produces an extreme result. Type II: mainly a lack of statistical power (for example, a sample that is too small).
2. Mostly significant results are published, so the literature contains more Type I errors and overstates relationships. Registered reports are judged on the question and method before the results are known, so a non-significant result is as publishable as a significant one.
3. 1 − .59 = .41.
4. 128 for the *t* test; 84 for the test of *r*.
5. 97 of the 100 original studies had significant effects, but only 36 replications did, and the replications' effect sizes were on average half as large.
6. Hypothesizing after the results are known: mining the data without an *a priori* hypothesis and then claiming that a significant result had been predicted, so a possible chance result looks like a confirmed prediction.

Part 2

1. (a) Concluding that one therapy is more effective than the other when in the population they are equally effective. (b) Concluding that there is no evidence of a difference when one therapy really is more effective (for example, because the samples were too small).
2. (example answer; other answers are possible) Benefits: pre-registration makes HARKing and *p*-hacking visible, because readers can compare the plan with the report; shared data and materials let others check the analyses and replicate; more null results become visible, reducing the file drawer problem. Drawbacks: extra time and work; participants' privacy must be protected when data are shared; researchers may report fewer unexpected findings (which can still be reported as exploratory).
3. (example answer) By changing the choices (which party, which politicians, which measure of the economy, whether to exclude recessions), you can usually find a combination that gives $p < .05$. Each choice looks reasonable on its own, which is why *p*-hacking is easy to do without noticing. Pre-registering the choices would prevent it.

Part 3 (example answers)

1. The failure to replicate does not necessarily mean that the original study was wrong.
2. Although null hypothesis testing is criticized, it is still the dominant approach.
3. The published literature is likely to overstate the strength of relationships.
4. Pre-registration may help to reduce the replicability crisis.

Writing task (example answer) "The hypotheses, the exclusion criterion and both analyses were pre-registered. One participant in the wake condition was excluded because she fell asleep during the rest period. As predicted, participants in the nap condition recalled significantly more word pairs (M = 22.00, SD = 2.54) than those in the wake condition (M = 17.00, SD = 3.09), $t(18) = 3.95$, $p < .001$, d =

1.86, 95% CI for the difference [2.34, 7.66]. In the nap condition, the positive correlation between minutes asleep and recall was not statistically significant, $r(8) = .45$, $p = .20$. Because only 10 participants contributed to this analysis, whereas about 28 are needed to detect a strong correlation with power of .80, a Type II error cannot be ruled out."

My notes